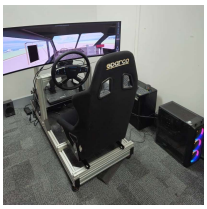
HGV Driver Simulator for Roll Stability Analysis and Driver Training

Reduced-order modelling and real-time rollover awareness for modified Rosenbauer Panther 8x8

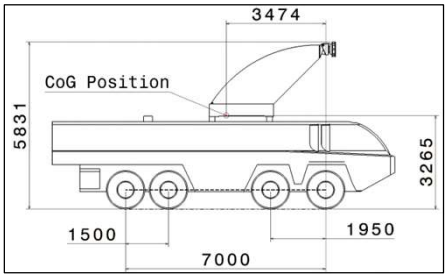


1) THE PROBLEM

Modified Rosenbauer Panther 8x8 carries a roof-mounted EVTOL propulsion test asset, raising vehicle mass and centre-of-gravity height. This reduces lateral acceleration limit before wheel lift.

Research question: Can a reduced-order, real-time model reproduce the vehicle's lateral and roll response and provide a continuous estimate of proximity to wheel unloading for driver-in-the-loop use?

28,828 kg Total vehicle mass	3.27 m Centre-of-gravity height
2.5 m Track Width	0.38 Static Stability Factor



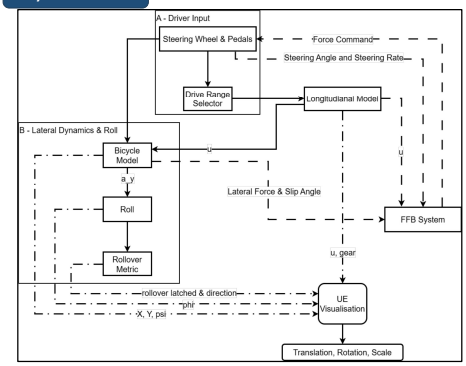
Sketch of vehicle with CoG estimate placement

2) AIM & OBJECIVES

Aim: Develop and evaluate a reduced-order DiL framework for lateral and roll response with continuous estimation of proximity to wheel lift

- **O1** Integrated model with a continuous rollover index
- **O2** Parameterise by source, classify confidence
- **O3** Verify, then validate on FTT test data
- **O4** Quantify sensitivity to mass and CoG position
- **O5** Real-time DiL and a defined validity envelope

3) MODEL



Flowchart of overall model structure

Four-axle single track lateral model, two front axles steered coupled to a single DoF roll dynamics model with point mass longitudinal model feeding lateral model.

Fidelity concentrated on states that govern rollover Parameters classified by provenance, uncertainties accounted via sensitivity analysis

Assumptions:

- Single-track
- Linear tyre model
- Constant suspension parameters
- Fixed steer ratio of axle 2 of 0.41 to axle 1
- Airfield road typically flat, no excitations, no banking
- Post-wheel lift fixed outer tyre contact line

6) DiL Implementation

Driver
Moza wheel + Pedals
P – R – N – D selector

→

Simulink
4-axle + 1 DoF roll model
ode3, fixed step 60 Hz

→

Unreal Engine
Sim 3D Actor pose
Airfield + HUD

Visual scene + HUD lamp at LTR ≥ 0.8

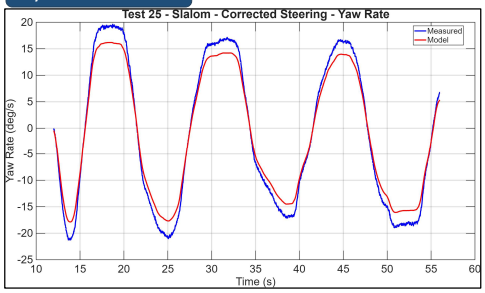
Cotswold Airport traced in Blender from satellite imagery (Google Earth).

Rendering is limiting factor for computational costing. Scalability on Unreal Engine set to low improves visual performance at cost of rendering fidelity. FPS at low > 60

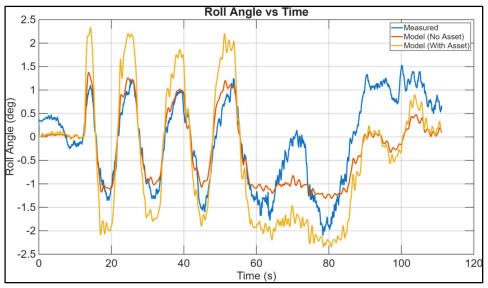


Cockpit view of truck with HUD display including warning light display

4) VALIDATION



Yaw-rate slalom segment, zeroed calibration (TEST 25)



Measured roll angle vs model (asset and no asset) (TEST 25)

Yaw-rate NRMSE against FTT data was 14.8% (slalom) and 16.5% (sustained cornering). Roll disagreement stemmed from pre-asset data produced in July 2024.

With configuration matched parameters; NRMSE fell to 36.6% for slalom and 35.7% on double lane change. Roll is more representative to bare-truck parameters.

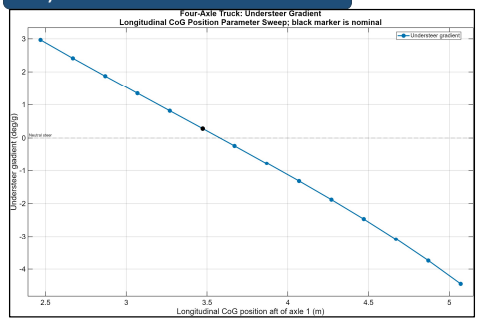
7) Conclusion

- 4-axle vehicle model implemented in real time coupled to DiL simulator
- Principle yaw-rate and roll-response trends, remaining error attributable to vehicle/configuration uncertainty
- CoG position governs handling balance and rollover proximity
- Continuous rollover-awareness feedback producible and useful for prediction of rollover from driver inputs

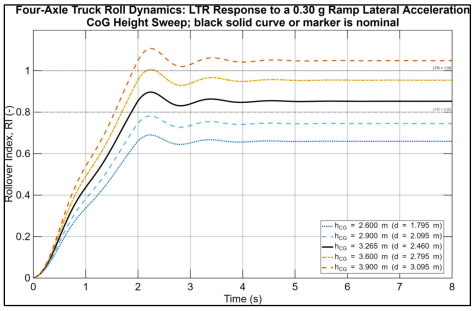
8) Future Work

- Roof-asset model validation with measured data with roof-asset included on Rosenbauer
- Include suspension compliance, non-linear roll characteristics and load-sensitive tyre behaviour
- Verify LTR threshold by driver feedback sessions
- Evaluate latency on simulator use possible with use of high-speed camera

5) ANALYSIS & SENSITIVITY



Understeer gradient as CoG position aft axle 1 is swept (2.47 → 5.07 m)



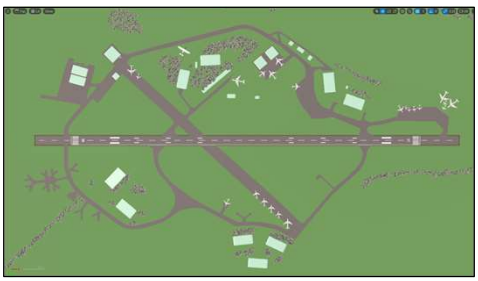
LTR as CoG height is swept (2.6 → 3.9 m)

Longitudinal CoG position dominates handling:
Understeer gradient moves from + 2.97 → -4.44 deg/g (2.47 → 5.07 m)

CoG height dominates rollover proximity:
LTR at 0.3g: 0.66 → 1.05 (2.6 → 3.9 m)

Nominal CoG (3.27 m) LTR reaches warning (LTR > 0.8) at 0.3g input

Near resonance, rollover threshold at nominal CoG height reduces from ~0.4g to ~0.2g



Map of Cotswold Airfield imported into Unreal Engine. Traced on Blender

References.

Gillespie (1992) Fundamentals of Vehicle Dynamics

Rajamani (2012) Vehicle Dynamics & Control

Babulal et al. (2015) J. Terramechanics

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